

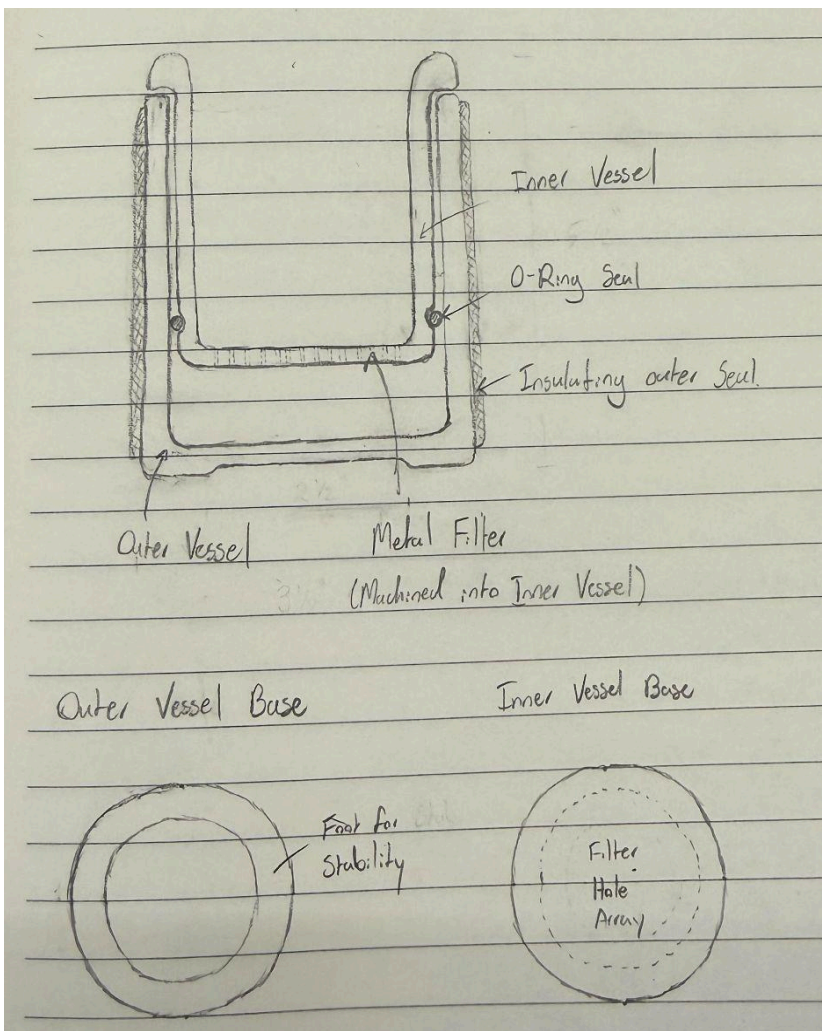
Project Approach Rationale

8 designs and additional parts sketches (see below) were developed to brainstorm possible approaches for The Hand Ram Coffee Press.

The final approach was determined to be Design 8: an integrated coffee press and cup.

Unlike the original design, the integrated coffee cup does not require a separate cup or mug to press the coffee into, eliminating the need to fit a standard mug. Furthermore, the integrated design is more stable than the original, where the user was required to put a substantial amount of force onto the press that sat on top of a cup that is of unknown condition. The design is also better suited for making coffee on-the-go, since the user does not need to carry a separate cup to use the press. Additionally, this design allows the user to not worry about cleaning a separate cup along with the coffee press. Lastly, the integrated design loses no functionality over the original design.

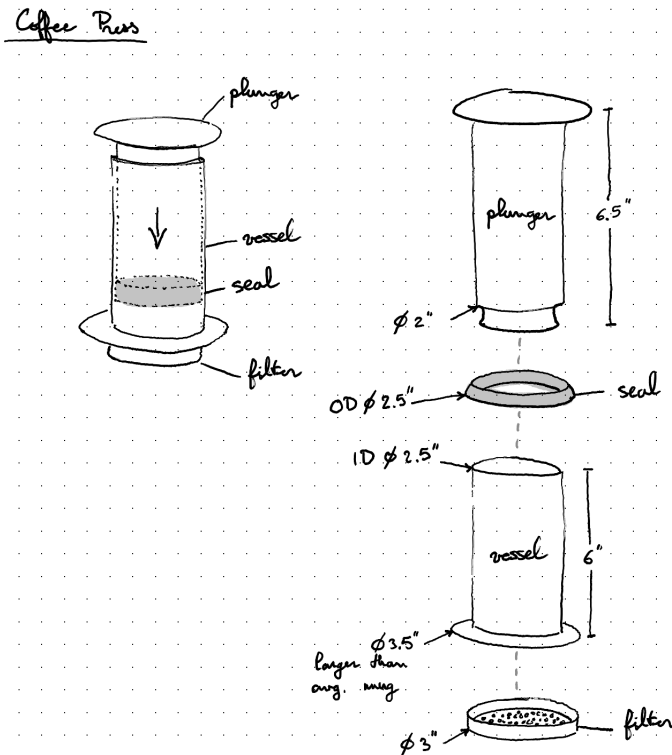
The following sketch was used to draft the CAD drawings included in DOC02:



All initial design sketches are below:

Design 1:

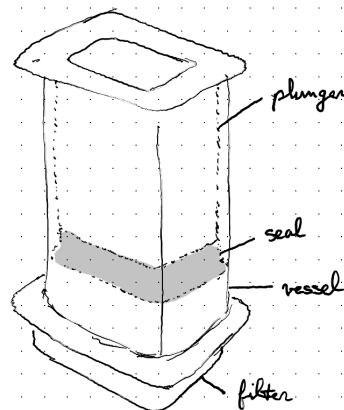
- **Concept:** Design based on Aeropress coffee press
- **Pros:** already made ... we know it works
- **Cons:** already made...can we improve it?



Design 2:

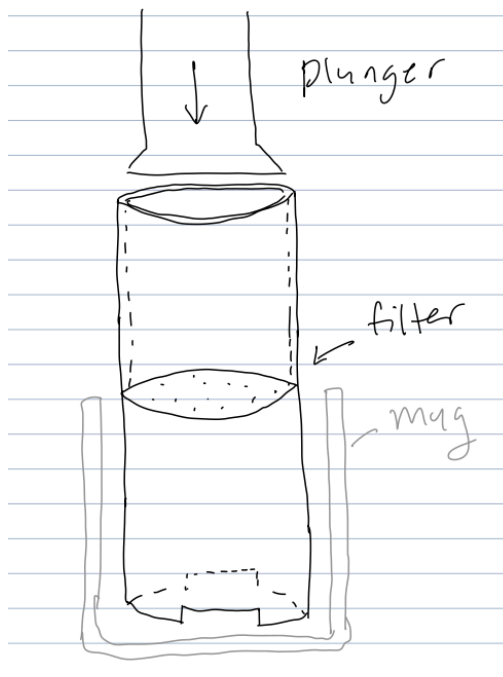
- **Concept:** A square version of the Aeropress coffee press
- **Pros:** idk man
- **Cons:** fitting on a circle, cleaning corners, seal at corners

Square Hand Ram!



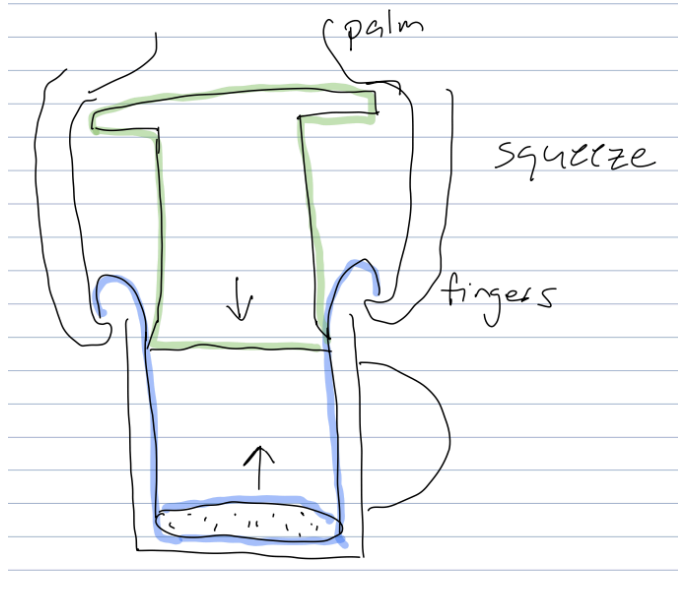
Design 3:

- **Concept:** Elongate vessel for improved stability (the press rests on the bottom of mug rather than the lip of the mug)
- **Pros:** increased stability/decreased risk of spilling boiling water
- **Cons:** more material, extension might need to be removable to access filter for cleaning



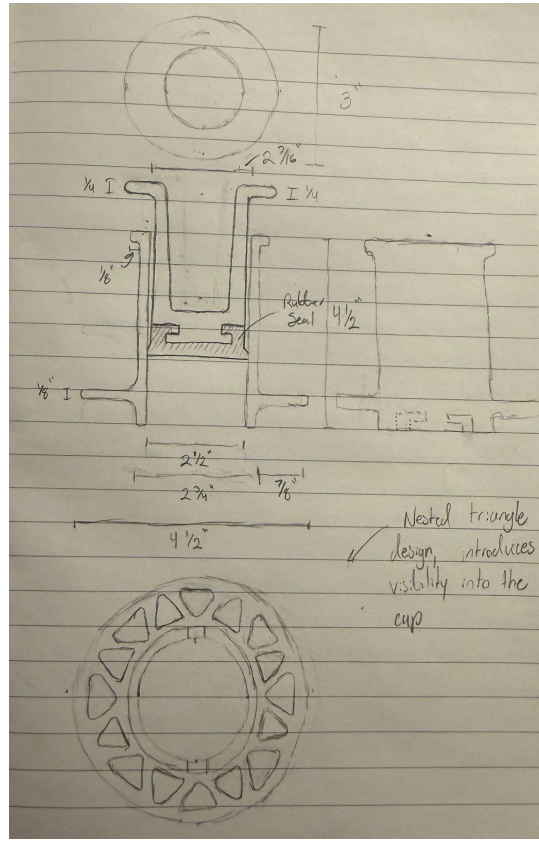
Design 4:

- **Concept:** Use of both hands to filter coffee grinds out. (A nice grip strength exercise)
- **Pros:** both hands can be used for pressing without need to stabilize mug
- **Cons:** size of product is limited by hand size



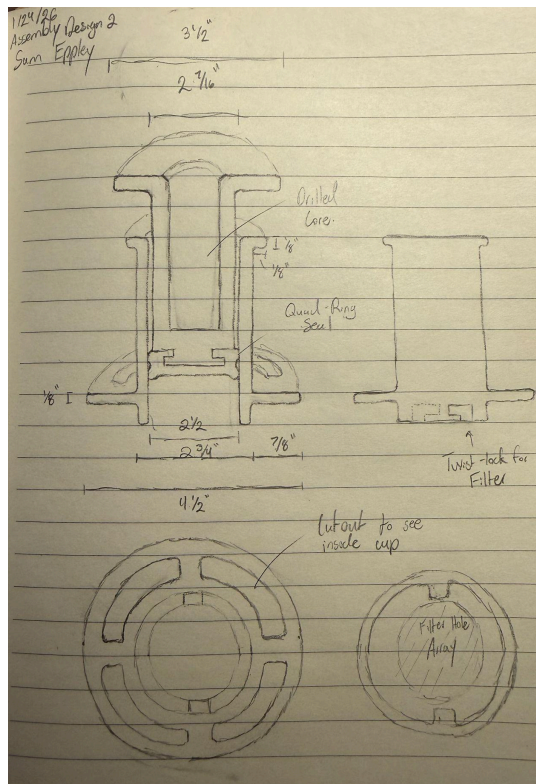
Design 5:

- **Concept:** Improved Aeropress
- **Pros:** allows air to escape as liquid is pressed into mug, visibility into the cup, potentially fewer material required
- **Cons:** more manufacturing processes to do cutouts, locking system for filter may be complicated



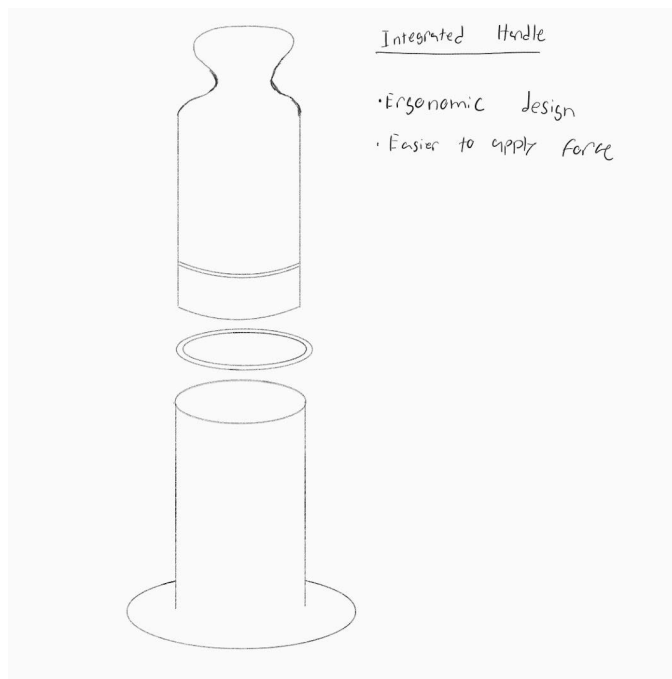
Design 6:

- **Concept:** Improved Aeropress
- **Pros:** allows air to escape as liquid is pressed into mug, visibility into the cup, potentially fewer material required, simpler cutout design than Design 5
- **Cons:** similar to Design 5



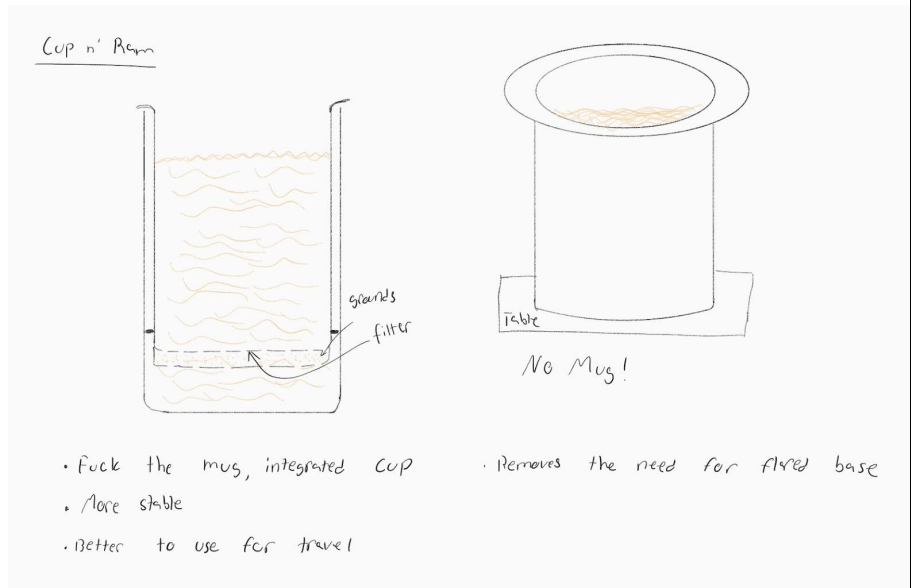
Design 7:

- **Concept:** Press with special handle
- **Pros:** see sketch
- **Cons:** hard to confirm if design is easier to apply force



Design 8:

- **Concept:** Single-serve french press design
- **Pros:** see sketch
- **Cons:** see sketch



Additional parts sketches:

